SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling. (BL4)

QUESTIONS: 05 **Total Marks:100**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I G SAI TEJA (192472137) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

RUBRICS FOR CALCULATION

TECHNICAL PRESENTATION					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Technical Presentation

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage. Explain how:

A. Python programming basics are used
B. Pandas and NumPy are used for data handling
C. Matplotlib and Seaborn are used for result visualization

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("D:\Downloads\student_data.csv")

print("Student Data")
print(df)

average_marks = np.mean(df["Marks"])
average_attendance = np.mean(df["Attendance"])

pass_students = df[df["Marks"] >= 50].shape[0]
total_students = len(df)

pass_percentage = (pass_students / total_students) * 100

print("\nAverage Marks =", average_marks)
print("Average Attendance =", average_attendance)
print("Pass Percentage =", pass_percentage)

plt.figure(figsize=(6,4))
plt.bar(df["Student"], df["Marks"])
plt.title("Student Marks")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()

plt.figure(figsize=(6,4))
sns.barplot(x="Student", y="Attendance", data=df)
plt.title("Student Attendance")
plt.show()
```

OUTPUT:

```
>
===== RESTART: C:/Users/vipla/OneDrive/Pictures/Documents/Desktop/CO2 A2.py ====

===== RESTART: C:/Users/vipla/OneDrive/Pictures/Documents/Desktop/CO2 A2.py ====
Student Data
  Student  Marks  Attendance
0    Arun     85          90
1    Bala     72          85
2  Charan     65          80
3  Deepak     91          95
4   Eswar     55          70
5  Farhan     78          88
6   Gokul     45          60
7    Hari     68          75

Average Marks = 69.875
Average Attendance = 80.375
Pass Percentage = 87.5
```